ABHISHEK SINGH

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SUMMARY

Software Engineer with 4+ years building scalable backend systems in Python, Django, and FastAPI, specializing in GenAI applications — RAG pipelines, local LLM deployment, and AI-driven data classification. Skilled at designing REST APIs, integrating databases, and shipping production features end to end. Seeking a backend/AI engineering role to build reliable, privacy-conscious AI products.

EXPERIENCE

Software Engineer Nov 2024 - Present
Iota Analytics *Mohali, India*

- Architected and implemented a secure RAG chatbot enabling users to query proprietary data while enforcing strict data privacy through local LLM deployment (Ollama).
- Engineered a scalable rule-based patent classification engine to automatically evaluate and score patent relevance across large datasets.
- Automated patent tagging workflows, reducing manual review effort by approximately 70% while improving classification consistency across 50,000+ patent records.
- Engineered and trained an AI-powered patent classification system on historical patent data to automatically tag and categorize new records.

Software Developer Aug 2023 - Sep 2024
Snakescript Solutions LLP *Mohali, India*

- Designed and delivered 100+ REST APIs in Django/DRF supporting authentication, reporting, and core business workflows for production applications.
- Implemented Celery-based asynchronous email notifications, automatically sending ticket confirmation emails upon ticket creation, improving application responsiveness and user communication.
- Built WebSocket-based real-time messaging and live data streaming features, supporting hundreds of concurrent client connections with low-latency communication.
- Collaborated with cross-functional teams to analyze requirements and deliver end-to-end solutions on schedule.

Associate Software Engineer Apr 2022 - Jul 2023
bebo Technologies *Chandigarh, India*

- Integrated the AI-powered resume parser with backend services and databases to enable efficient storage, retrieval, and processing of candidate data, streamlining hiring workflows and contributing to a 30% increase in overall productivity.
- Applied **Object-Oriented Programming (OOP)** principles and design patterns to develop scalable, maintainable, and high-quality software solutions.
- Optimized SQL queries and backend services, reducing API response latency by approximately 40% and improving overall application performance.

PROJECTS

Secure RAG Chatbot with Local LLMs. Designed and architected a secure Retrieval-Augmented Generation (RAG) chatbot enabling users to query and interact with their private datasets. Implemented local LLM inference using Ollama to ensure complete data privacy by eliminating external API calls and preventing data exposure. Built document ingestion pipelines, vector indexing, and contextual retrieval mechanisms to deliver accurate, relevant, and context-aware responses based exclusively on user-owned data.

AI-Powered Patent Classification & Rule Engine. Developed an AI-powered Patent Classification and Rule Engine using Python and Django to automate the analysis of large-scale patent datasets. Built a configurable rule engine dashboard that enabled users to create custom classification rules without code changes. Leveraged AI-driven content analysis, keyword matching, and rule-based logic to improve classification accuracy, reduce manual review efforts, and efficiently process high-volume patent data.

E-commerce Website. Developed and enhanced the backend architecture of a scalable e-commerce platform using Python and Django, implementing core features such as product catalog management, shopping cart, order processing, and secure checkout. Devised and integrated a dynamic promotions engine to support real-time discounts, coupons, and pricing rules. Built and optimized RESTful APIs and database operations to improve application performance, transaction reliability, and scalability, delivering a seamless and efficient online shopping experience. (Try it here)

Home Healthcare Platform (Web + Mobile). Developed a Home Healthcare Web and Mobile Platform by building a Django-based backend with Django REST Framework to support both web and cross-platform mobile applications (iOS and Android). Implemented RESTful APIs for appointment scheduling, user management, service requests, and real-time communication between patients and healthcare professionals. Integrated an AI-assisted chatbot for automated user support and query resolution, while optimizing API performance, security, and scalability to deliver a seamless healthcare experience across web and mobile platforms. (Try it here)

Real-Time Chat Application. Developed a real-time chat application using FastAPI, enabling instant direct messaging with WebSocket-based communication for low-latency message delivery. Implemented secure user authentication, online/offline presence tracking, message persistence, and chat history using RESTful APIs and a relational database. Designed a scalable backend architecture with asynchronous processing to support concurrent users and deliver a seamless real-time messaging experience similar to modern chat platforms.

AWS Lex. Integrated an AWS Lex chatbot to provide instant user support and answer common healthcare queries. Enabled real-time communication with healthcare professionals, improving user engagement, response time, and overall service efficiency.

TECHNICAL SKILLS

Languages	Python, JavaScript, HTML, CSS
Backend & Web Frameworks	Flask, Django, Django REST Framework (DRF), FastAPI
APIs & Architecture	RESTful APIs, OOP, Microservices Architecture
Data & Visualization	Pandas, NumPy, Matplotlib
Databases	PostgreSQL, MySQL, Graph Databases
Tools & DevOps	Docker, Redis, CI/CD, Git, Pytest, Selenium, AWS (EC2, S3 – basic)
AI / ML & GenAI	Vector Database (FAISS), Embedding/Semantic Search, LangChain, NLP, RAG, LLM, Prompt Engineering

SOFT SKILLS

Problem-solving, Continuous Learning, Teamwork, Communication

EDUCATION

Bachelor of Technology in Computer Science Engineering, CT Group of Institutions	2018 – 2022
Senior Secondary (XII), D.A.V Senior Secondary School	2017 – 2018
Secondary (X), Ch. Balbir Singh Public School,	2015 – 2016

EXTRA-CURRICULAR ACTIVITIES

- **Leadership Role:** Managed teams in the absence of team leaders, ensuring successful project completion and timely delivery.
- **Continuous Learning:** Actively explore Generative AI, LLMs, LangChain, RAG, and FastAPI through hands-on projects and continuous technical learning.
- **Competitions and Awards:** Silver Medal, Abacus Competition – Awarded for excellence in abacus skills, demonstrating precision and analytical thinking.